Seongmin Jung

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Research Interest

Multimodal Robot Learning, Vision-Language(-Action) Models, 3D Computer Vision, Embodied AI

Education

Seoul National University — Sep 2024 – Aug 2026 (Expected)
M.S. in Artificial Intelligence (Advisor: Prof. Jongwoo Lim) — Seoul, South Korea
GPA: 4.25/4.30

Yonsei University — Mar 2019 – Feb 2024
B.S. in Electrical and Electronic Engineering — Seoul, South Korea
GPA: 4.02/4.30 • Graduated with **High Honors (Top 3% in college)**

Case Western Reserve University — Aug 2022 – May 2023
Exchange Program — Cleveland, OH, USA

Honors and Awards

The National Scholarship for Science and Engineering — 2019 – 2024
Korea Student Aid Foundation (KOSAF) — Seoul, South Korea
- Full merit-based scholarship awarded to top STEM students by the South Korean government, covering full tuition.

High Honors — Fall 2019
Dept. of Electrical & Electronic Engineering, Yonsei University — Seoul, South Korea

Highest Honors — Spring 2019
Dept. of Electrical & Electronic Engineering, Yonsei University — Seoul, South Korea

Preprints & Manuscripts

[1] **Seongmin Jung***, Seongho Choi*, Gunwoo Jeon, Minsu Cho, and Jongwoo Lim. PanoGrounder: Bridging 2D and 3D with Panoramic Scene Representations for VLM-based 3D Visual Grounding. *ECCV 2026, under review.* ArXiv: https://arxiv.org/abs/2512.20907

Research Experiences

AI4CE Lab, *New York University*
Internship (Advisor: Prof. Chen Feng) — Remote / New York City, NY, USA
Tactile-Informed Imitation Learning — Nov 2025 – Present
- Developing visual–tactile compositional diffusion policies that fuse low-rate visual inputs with high-rate piezoelectric tactile signals for contact-rich manipulation.
- Constructed a LeRobot-based training pipeline, reimplementing baselines for integration with the custom robot hardware.

Robot Vision Lab, *Seoul National University*
Graduate Research — Seoul, South Korea
Solving Rubik's Cube with Bimanual Robot Hands — Sep 2025 - Present
- Collected VR teleoperation data on a Fourier GR-1 bimanual humanoid robot in Isaac Lab using a Meta Quest 3.
- Fine-tuned an open-source VLA (GR00T N1.5) for stable grasping and rotation of Rubik's Cube faces.

PanoGrounder — Jan 2025 – Mar 2026
- Proposed a framework for VLMs reasoning over 3D Mesh/3DGS via panoramic rendering, enabling spatial understanding.
- Fine-tuned a ∼17B VLM on 8x A100 GPUs using DeepSpeed/LoRA, injecting DINOv2 features via custom adapters.
- Achieved SOTA (+4.7% on Nr3D), robust generalization (+17.4% on ARKitScenes), and efficient inference (∼3.3s/query).

Neural SLAM with Point Cloud Projection for Faster RGB-D Tracking — Sep 2024 - Dec 2024
- Proposed a point-NeRF-based differentiable camera pose tracker, replacing iterative NeRF queries along camera rays.
- Replaced the tracking module of Point-SLAM, achieving over 3× faster tracking while preserving trajectory accuracy.

Biological Cybernetics Lab, *Yonsei University*
Undergraduate Thesis (Advisor: Prof. DaeEun Kim) — Seoul, South Korea
Autonomous Navigation for High-Speed Ground Robot — Jul 2023 – Dec 2023
- Implemented low-latency obstacle avoidance and path planning in C++, using ROS and LiDAR odometry.
- Deployed on a Raspberry-Pi-powered RC car achieving high-speed navigation at max 5m/s.

Professional Experience

Stealth Mode MedTech Startup Oct 2025 – Present
3D Vision Engineer Remote / Pittsburgh, PA, USA

- Developing 3D volumetric segmentation networks (e.g., 3D U-Net) for processing noisy, high-resolution voxel data.
- Building data workflow, 5-fold cross-validation, and clinical evaluation.
- Collaborating with medical professionals to define clinical problems and enhance performance for real-world deployment.

Mobiltech Aug 2023 – Jun 2024
Research Assistant Seoul, South Korea

Enhancing Pose Tracking in Urban Scenes

- Developed a LiDAR odometry pipeline in C++, using per-frame height maps of surrounding structures, improving trajectory accuracy and robustness in dense urban scenes.

Isaac Sim Environment for Heterogeneous Multi-Robot Systems

- Implemented a heterogeneous multi-robot simulation with ground and aerial platforms in Isaac Sim, integrating per-robot control and sensor setups in ROS.

Teaching Experiences

Digital Computer Concept and Practice (F37.202), *Seoul National University* Spring 2025, Spring 2026
Teaching Assistant Seoul, South Korea

- Led weekly C++ programming lab sessions and developed course materials and assignments.

Patents

[1] Y. Lee, **S. Jung**, J. Jo, and Y. Kim. 2023. Apparatus for collecting cigarette butts with IoT. Korea Patent 1028688440000, September 30, 2025.

Club Activities

CWRUbotix, *Robotics Club at Case Western Reserve University* Dec 2022 – Jun 2023
MATE ROV World Championship 2023 Cleveland, OH, USA

- Constructed underwater robot simulation for mission environment using ROS and Gazebo Simulation.
- Designed and implemented an Arduino-based wireless communication module for mission execution.

VERY, *Entrepreneurship Club at Yonsei University* Mar 2021 – Feb 2022
Team Lead Seoul, South Korea

- Established online music ensemble platform based on self-recording for college bands.
- Led a 4-member team, coordinating operation, design, and development throughout the project.

Leadership

Korean Mentor Sep 2023 – Aug 2024
Office of International Affairs, Yonsei University Seoul, South Korea

- Mentored 10 incoming study-abroad students, aiding in cultural and academic adaptation in Korea.

President of Student Council Nov 2020 – Oct 2021
Dept. of Electrical & Electrical Engineering, Yonsei University Seoul, South Korea

- Led key events including freshmen orientation and homecoming, organizing 300+ students.

Technical Skills

Programming Languages: Python, C/C++, MATLAB, Verilog

Libraries & Frameworks: PyTorch, DeepSpeed, LeRobot, OpenCV, Open3D, 3DGS, NeRF

Robotics & Simulation: ROS 1 & 2, Isaac Sim, Isaac Lab, Gazebo, VR Teleoperation (Meta Quest3 + XR Robotics)

Languages: Korean (native), English (fluent)

- TOEFL iBT: 108 (Reading: 30, Listening: 29, Speaking: 24, Writing: 25)